

TRADITIONAL BAYES AND THE BAYESIAN METHOD OF MOMENT ANALYSIS FOR THE MIXED LINEAR MODEL WITH AN APPLICATION TO ANIMAL BREEDING

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Summary: The Bayesian Method of Moments (BMOM) was introduced by Arnold Zellner in 1994. Given the data it enables researchers to make inverse probability statements about unknown parameters if the form of the likelihood function is unknown. In this paper the theory and results derived by Zellner (1972a) are extended to the mixed linear model. To illustrate the application of the method, data from the Elsenburg Dormer sheep stud were analysed. A total of 879 weaning weight records, from the progeny of 17 sires were used. The estimates obtained were compared with the traditional Bayes and REML estimates.

1. Introduction

The Bayesian Method of Moments (BMOM) was introduced by Arnold Zellner in 1994. Given the data, it enables researchers to compute post data densities for parameters and future observations if the form of the likelihood function is unknown. The BMOM approach provides a solution to the famous inverse problem posed by Bayes (1763) and hence the name Bayesian Method of Moments.

An essential element of the Bayesian approach is Bayes' theorem, also referred to in the literature as the principle of inverse probability. In problems involving "inverse probability" we have given data and from the information in the data try to infer what random process generated them. On the other hand, in problems of "direct probability" we know the random process, including values of its parameters and from this knowledge make probability statements about outcomes or data produced by the known random process. Problems of statistical estimation are thus seen to be problems in "inverse probability", whereas many gambling problems are problems in direct probability.

BMOM has been applied to multiple and multivariate regression models as well as simultaneous equation problems. For further details see Zellner (1995, 1996, 1997a, 1997b), Zellner, Tobias and Ryu (1999), Zellner and Sacks (1996), Tobias and Zellner (1997), Currie (1996) and Zellner et al. (1994). See also Green and Strawderman (1996) for a study in which the authors did not have enough information to formulate a satisfactory likelihood function, thus they made use of BMOM.

In the BMOM approach the posterior and predictive moments, based on a few relatively weak assumptions are used to obtain maximum entropy densities for parameters, realized error terms and future values of variables. Shannon

(1948) defines entropy (or uncertainty) as

$$W = - \int p(y) \log p(y) dy$$

where $p(y)$ is a probability density function.

Maximizing W subject to various side conditions is well known in the literature as a method for deriving the forms of minimal information distributions.

Shannon (1948) has also indicated how maximum entropy (ME) distributions can be derived by a straightforward application of calculus of variations techniques. In particular, he has shown that the ME distribution that maximizes entropy subject to a normalization condition is just the uniform distribution. By adding additional side conditions giving the first two moments of the distribution are imposed, the ME distribution is the normal distribution. On the other hand if just the side conditions on the zeroth and first moments are utilized, the maxent density is an exponential density.

For discussion and application of maximum entropy, see e.g. Jaynes (1982, 1988), Shore and Johnson (1980), Caver and Thomas (1991) and Zellner and Highfield (1988). Zellner (1997a) also discussed coherent procedures of updating BMOM maxent post-data densities for parameters and future observations.

In this paper the theory and results derived by Zellner (1997a) will be extended to the mixed linear model.

2. Description of the Practical Problem and the Classical Frequentist Solution

The data used in this study were from the Dormer sheep stud started at the Elsenburg College of Agriculture near Stellenbosch in 1940. The main object in developing the Dormer was the establishment of a mutton sheep breed which would be well adapted to the conditions prevailing in the Western Cape (winter rainfall) and which could produce the desired type of ram for crossbreeding purposes (Swart 1967).

Single sire mating was practised with 25 to 30 ewes allocated to each ram. A spring breeding season (6 weeks duration) was used throughout the study. Season therefore had to be included as a fixed effect as a birth year–season concatenation.

During lambing the ewes were inspected daily and dam and sire numbers, date of birth, birth weight, age of dam, birth status (type of birth) and size of lamb recorded. The same procedure was repeated every two weeks until all the lambs were weaned.

The elements of the data vector therefore represent the weaning weights of lambs in kilogram. These lambs were born in the period 1943–1950. A total of 879 weaning weight records, from the progeny of 17 sires were available after editing.

It is clear that only a small sample from the Elsenburg Dormer stud was used. It was not our intention to reanalyse the data from a genetic point of view, rather, we used it to illustrate how the Bayesian approach and Gibbs sampling could be applied to real animal breeding problems. Whenever appropriate, comparisons will be drawn in Section 6 between the Bayes estimates (using Gibbs sampling) obtained from MATLAB and written by us, as well as the estimates calculated

using Restricted Maximum Likelihood (REML) and the Bayesian Method of Moments (BMOM) estimates. The classical (REML) estimates were obtained by using the MTDFREML program developed by Boldman *et al.* (1995).

In animal breeding applications, it is usually assumed that the data follows a mixed linear model. The analysis was therefore carried out by using such a model; in this case known as a sire model. The mixed linear model postulates that the observable random vector $\underline{y}$ ($n \times 1$) is a linear combination of fixed and random effects plus a random residual error (Gianola and Fernando, 1986). In its simplest form the univariate mixed linear model can be written in matrix notation as

$$\underline{y} = X\underline{\beta} + Z\underline{u} + \underline{\varepsilon} \quad (2.1)$$

Here

- (i) $\underline{y}$ is an $n \times 1$ data vector;
- (ii) X is a known incidence matrix of order $n \times p$;
- (iii) $\underline{\beta}$ is a $p \times 1$ vector of unknown parameters, usually called the fixed effects and uniquely defined so that X has full column rank p .
- (iv) $\underline{u}$ is a $q \times 1$ vector of unobservable random effects. The distribution of $\underline{u}$ is usually considered to be normal with mean vector $\underline{0}$ (the null vector of order $q \times 1$) and variance-covariance matrix $\sigma_u^2 A$ (where A represents a $q \times q$ known matrix and σ_u^2 an unknown scalar). In a Bayesian analysis the distribution of $\underline{u}$ is called the prior distribution of $\underline{u}$.
- (v) Z is a known, fixed matrix of order $n \times q$;
- (vi) $\underline{\varepsilon}$ is an $n \times 1$ unobservable vector of random residuals such that the distribution of $\underline{\varepsilon}$ is n -dimensional normal with mean vector $\underline{0}$ and variance-covariance matrix $\sigma_\varepsilon^2 I_n$ (where I_n represents an $n \times n$ identity matrix and σ_ε^2 an unknown scalar);
- (vii) the vectors $\underline{\varepsilon}$ and $\underline{u}$ are statistically independent;
- (viii) σ_u^2 and σ_ε^2 are unknown scalar-value parameters called the variance components.

As mentioned, the elements of $\underline{y}$ represent the weaning weights of lambs in kilogram; and the q elements of $\underline{u}$ represent the breeding values (or transmitting abilities) of the sires (male parents). Therefore in the case of a sire model, a breeding value refers to a sire's value as a parent in a breeding program. It is a measure of the animal's progeny performance relative to the mean value of its breed. In this case, the $q \times q$ matrix A is called the relationship (genetic covariance) matrix and its elements reflect the genetic relationships among the q sires (see e.g. Henderson 1974). The data as well as the A matrix are given in Hugo (1998).

For our example $\underline{\beta} (p \times 1) = (\beta_0, \underline{\beta}'_1, \dots, \underline{\beta}'_4)'$

where $\underline{\beta}'_1 = (\beta_1, \beta_2, \dots, \beta_7) :$ Year (season of birth) effects;

$\underline{\beta}'_2 = (\beta_8, \beta_9, \dots, \beta_{13}) :$ Age of dam effects;

$\underline{\beta}'_3 = \beta_{14} ;$ Sex of lamb effect;

and $\underline{\beta}'_4 = (\beta_{15}, \beta_{16}) :$ Birth status effects.

The years were from 1943–1950, age of dams from 2 to 8 years and older, sex of lamb was male and female and birth status was individual, twins and triplets. These fixed effects are therefore the effects attributable to a finite set of levels of a factor which occur in the data and which are there because we are interested in them. According to Gianola and Fernando (1986) a “fixed” effect is in a Bayesian sense, a random variable on which prior knowledge is diffuse or vague.

Random effects on the other hand are attributable for a (usually) infinite set of levels of a factor, of which only a random sample are deemed to occur in the data. Usually a proper (normal) prior is associated with a random effect.

The classical frequentist solution of the mixed model can be obtained from Henderson's mixed model equations. Henderson, in Henderson *et al.* (1959), developed a set of equations that simultaneously yield best linear unbiased predictors of the random effects and best linear unbiased estimators of the fixed effects. They were derived by maximizing the joint density of $\underline{y}$ and $\underline{u}$; i.e.

$$f(\underline{y}, \underline{u}) = \left(\frac{1}{2\pi\sigma_\varepsilon^2} \right)^{n/2} \exp \left\{ -\frac{1}{2\sigma_\varepsilon^2} (\underline{y} - X\underline{\beta} - Z\underline{u})' (\underline{y} - X\underline{\beta} - Z\underline{u}) \right\} \\ \times \left(\frac{1}{2\pi\sigma_u^2} \right)^{q/2} |A|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2\sigma_u^2} \underline{u}' A^{-1} \underline{u} \right\}. \quad (2.2)$$

Equating to zero the partial derivatives of (2.2) with respect to elements first of $\underline{\beta}$ and then of $\underline{u}$ give the mixed model equations, written more compactly as

$$\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + \frac{\sigma_\varepsilon^2}{\sigma_u^2} A^{-1} \end{bmatrix} \begin{bmatrix} \underline{\tilde{\beta}} \\ \underline{\tilde{u}} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix} \quad (2.3)$$

where $\underline{\tilde{\beta}}$ and $\underline{\tilde{u}}$ denote the solutions of $\underline{\beta}$ and $\underline{u}$. The classical approach to the problem of predicting linear combinations of fixed and random effects, when the values of the variance components are not known, has been to estimate the variance components and to proceed thereafter as if these estimates were the true values.

Patterson and Thompson (1971) developed a method to derive unbiased estimates of the unknown variance components σ_u^2 and σ_ε^2 based on the maximum likelihood principle; it is called restricted maximum likelihood estimation (REML). The restricted maximum likelihood method is based on the likelihood of a vector whose components are independent linear combinations of the observations. The basic idea is to end up with a random

vector that contains all the information on the variance components but no longer contains information on the fixed effects parameters.

Substituting the REML estimates of σ_u^2 and σ_e^2 in (2.3) yields the classical frequentist solution to the mixed linear model. For further information on these methods see for example Searle (1971, 1979), Henderson (1984) and Clarke (1998).

3. Bayesian Analysis of Mixed Linear Models via Gibbs Sampling

In Section 2 the classical approach to the problem of predicting linear combinations of fixed and random effects in the case of the mixed linear model was discussed. There are several problems with this approach.

- (i) The properties of the predictors are hard to assess.
- (ii) When the values of the variance components are estimated from the data, their sampling errors are generally not taken into account in the subsequent analysis. Therefore, the variance of the prediction error will generally be underestimated.
- (iii) Depending upon the size and characteristics of the data, point estimators of variance components can be highly variable. For certain values of the variance component estimators, the predictors obtained by substituting these values in the "Best Linear Unbiased Predictor" are intuitively unappealing.

An alternative approach to the classical approach is the Bayesian approach. Harville (1990), see also Gianola and Foulley (1990) stated "A more extensive use of Bayesian ideas by animal breeders and other practitioners is desirable and is more feasible from a computational standpoint than commonly thought. The Bayesian approach can be used to devise prediction procedures that are more

sensible – from both a Bayesian and Frequentist perspective – than those in current use”.

Some of the advantages of the Bayesian approach are:

- (i) The Bayesian practitioner does not need to commit himself to only a point estimate of the variance components. Credibility intervals can easily be obtained.
- (ii) Uncertainty about the true values of the variance components is formally incorporated into the analysis through the choice of an appropriate prior distribution.
- (iii) Given the data, the prior information about the unknown parameters and a well defined loss function, there exists an optimal Bayes predictor.
- (iv) All the available information about the random variable to be predicted is contained in the posterior distribution of the random variable. The practitioner can, therefore, base all of his inferences on this distribution.
- (v) The Bayesian approach is conceptually more appealing than the classical approach.

Critics of the Bayesian approach have most often cited the following points:

1. The Bayesian practitioner must formally express his prior beliefs about the unknown parameters in the form of a probability distribution.
2. The Bayesian methodology is computer intensive. In many situations, integration in several dimensions is required to obtain the required posterior distributions.

These might have been valid criticisms in the past but by using (a) Noninformative priors such as Jeffreys and Reference priors, and (b) Numerical Integration techniques such as Markov Chain Monte Carlo Methods and more specifically Gibbs Sampling, these problems can be overcome.

As discussed in Section 2 the mixed linear model postulates that the observable random vector $\underline{y}$ ($n \times 1$) is a linear combination of fixed and random effects plus a random residual error. In its simplest form the univariate mixed linear model can be written in matrix notation as

$$\underline{y} = X\underline{\beta} + Z\underline{u} + \underline{\varepsilon} \quad (3.1)$$

In the classical approach to analysis of data using a mixed linear model the distinctions of fixed versus random, known versus unknown, parameter versus statistic are all important. These classifications dictate the type of estimation and inference that is possible. When analyzing the mixed linear model (or any model) using a Bayesian approach, however, it only matters whether a specified quantity is observable or unobservable.

In equation (3.1) $\underline{y}$, X and Z are observable (because we see their values) while $\underline{\beta}$, $\underline{u}$ and $\underline{\varepsilon}$ are unobservable (because we do not see their values). No further classifications are necessary. In particular, both fixed effects and random effects are handled within the same general framework. In Bayesian modelling we treat $\underline{\beta}$, $\underline{u}$ and all variance components in the same way: they are unobservable.

For modelling the hierarchy, the distribution of $\underline{\varepsilon}$ gives the sampling distribution, which, in classical statistics, is the distribution of the data conditional on all the parameters. In a Bayesian analysis this distribution is called the likelihood function and it is always the first stage in a traditional Bayesian analysis with prior distributions relegated to other stages.

Therefore from (3.1) it follows that the conditional distribution which generates the data (likelihood function) is

$$\underline{y} | \underline{\beta}, \underline{u}, \sigma_{\varepsilon}^2 \sim N_n (X\underline{\beta} + Z\underline{u}, I_n \sigma_{\varepsilon}^2) \quad (3.2)$$

when I_s is the $s \times s$ identity matrix and $N_s(\underline{\mu}, \Sigma)$ denotes the s -dimensional multivariate normal distribution with mean $\underline{\mu}$ and variance-covariance matrix Σ .

As mentioned, an integral part of a traditional Bayesian analysis is the assignment of prior distributions to the unknown parameters in the model. The information contained in the prior distribution is combined with the information supplied by the data, through the likelihood function (if it is known), into the conditional distribution of the parameters given the data, which is known as the posterior distribution. Inferences about the unknown parameters are based on the posterior distribution (Gianola and Fernando 1986).

In the above model and for our example, "flat" or uniform prior distributions are assigned to σ_ϵ^2 and $\underline{\beta}$ as to represent lack of prior knowledge about the vector of fixed effects $\underline{\beta}$ and the error variance σ_ϵ^2 .

Therefore

$$p(\underline{\beta}, \sigma_\epsilon^2) = p(\underline{\beta}) p(\sigma_\epsilon^2) \propto \text{constant.} \quad (3.3)$$

Further, the prior distribution of the vector $\underline{u}$ is given by

$$\underline{u} | A, \sigma_u^2 \sim N_q(\underline{0}, A\sigma_u^2) \quad (3.4)$$

where A is a known matrix and σ_u^2 an unknown variance component. As mentioned in the case of a sire model, A is the relationship (genetic covariance) matrix of the sires and $\underline{u}$ is the vector of breeding values.

Also

$$p(\sigma_u^2) \propto \text{constant.} \quad (3.5)$$

The joint posterior distribution of the unknowns $(\underline{\beta}, \underline{u}, \sigma_u^2 \text{ and } \sigma_\epsilon^2)$ is proportional to the product of the likelihood function and the joint prior

distribution and is given by

$$p(\underline{\beta}, \underline{u}, \sigma_u^2, \sigma_\varepsilon^2 | D) \propto \left(\frac{1}{\sigma_\varepsilon^2} \right)^{n/2} \exp \left\{ -\frac{1}{2\sigma_\varepsilon^2} (\underline{y} - X\underline{\beta} - Z\underline{u})' (\underline{y} - X\underline{\beta} - Z\underline{u}) \right\} \\ \times \left(\frac{1}{\sigma_u^2} \right)^{q/2} \exp \left\{ -\frac{1}{2\sigma_u^2} \underline{u}' A^{-1} \underline{u} \right\} \quad (3.6)$$

where $D = (\underline{y}, X)$ denotes the data.

In many Bayesian problems, marginal posterior distributions are often needed to make appropriate inferences. However, due to the complexity of the joint posterior distribution, it is impossible to obtain these marginal densities analytically and because of the many unknowns it is very difficult to calculate numerically.

Recently due to work of Gelfand and Smith (1990), Gelfand *et al.* (1990), Carlin *et al.* (1992) and Gelfand *et al.* (1992), the Gibbs sampler has been shown as a useful tool for applied Bayesian inference in a broad variety of statistical problems. The Gibbs sampler is implicit in the work of Hastings (1970) and was made popular in the image processing context by Geman and Geman (1984). The Gibbs sampler is an adaptive Monte Carlo integration technique. The typical objective of the sampler is to collect a sufficiently large enough number of parameter realizations from conditional posterior densities in order to obtain accurate estimates of the marginal posterior densities. The principal requirement of the sampler is that all conditional densities must be available in the sense that random variates can be generated from them. The method is of great appeal on account of its simple logical foundation and reasonable ease of implementation.

The full conditional posterior density of each of the unknowns is obtained by regarding all other parameters in (3.6) as known. The required conditional

for $\underline{\beta}$ is

$$\underline{\beta} | \underline{u}, \sigma_u^2, \sigma_\varepsilon^2, D \sim N_p \left\{ \hat{\underline{\beta}}, (X'X)^{-1} \sigma_\varepsilon^2 \right\} \quad (3.7)$$

where $\hat{\underline{\beta}} = (X'X)^{-1} X'(\underline{y} - Z\underline{u})$. Note that this distribution does not depend on σ_u^2 .

The full conditional distribution of $\underline{u}$ is multivariate normal:

$$\underline{u} | \underline{\beta}, \sigma_u^2, \sigma_\varepsilon^2, D \sim N_q \left\{ \hat{\underline{u}}, \left(Z'Z + A^{-1} \frac{\sigma_\varepsilon^2}{\sigma_u^2} \right)^{-1} \sigma_\varepsilon^2 \right\} \quad (3.8)$$

where

$$\hat{\underline{u}} = \left(Z'Z + A^{-1} \frac{\sigma_\varepsilon^2}{\sigma_u^2} \right)^{-1} Z'(\underline{y} - X\underline{\beta}).$$

For the variance components the conditionals are

$$\begin{aligned} p(\sigma_\varepsilon^2 | \underline{\beta}, \underline{u}, \sigma_u^2, D) &= K_\varepsilon (\sigma_\varepsilon^2)^{-n/2} \\ &\times \exp \left\{ -\frac{1}{2\sigma_\varepsilon^2} (\underline{y} - X\underline{\beta} - Z\underline{u})' (\underline{y} - X\underline{\beta} - Z\underline{u}) \right\} \sigma_\varepsilon^2 > 0 \end{aligned} \quad (3.9)$$

an inverse gamma density where

$$K_\varepsilon = \left\{ \frac{(\underline{y} - X\underline{\beta} - Z\underline{u})' (\underline{y} - X\underline{\beta} - Z\underline{u})}{2} \right\}^{\frac{1}{2}(n-2)} \frac{1}{\Gamma\left(\frac{n-2}{2}\right)}$$

and

$$\begin{aligned} p(\sigma_u^2 | \underline{\beta}, \underline{u}, \sigma_\varepsilon^2, D) &= K_u \left(\frac{1}{\sigma_u^2} \right)^{q/2} \\ &\times \exp \left\{ -\frac{1}{2\sigma_u^2} \underline{u}' A^{-1} \underline{u} \right\} \quad \sigma_u^2 > 0 \end{aligned} \quad (3.10)$$

where

$$K_u = \left\{ \frac{u' A^{-1} u}{2} \right\}^{\frac{1}{2}(q-2)} \frac{1}{\Gamma\left(\frac{q-2}{2}\right)}.$$

We are often interested in the posterior distributions of functions of variance components like for example the intraclass correlation coefficient $\rho = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2}$ and the variance ratio $v = \frac{\sigma_u^2}{\sigma_\varepsilon^2}$. The conditional posterior distributions of these parameters can easily be obtained by making use of the theory of transformation of random variables.

For example the conditional posterior distribution of ρ can be obtained by using the transformation of $\sigma_u^2 \rightarrow \rho$ in the conditional posterior density of σ_u^2 (equation 3.10). Since the Jacobian of the transformation, $\frac{d\sigma_u^2}{d\rho} = \frac{\sigma_\varepsilon^2}{(1-\rho)^2}$, equation (3.10) can be written as

$$p(\rho | \underline{\beta}, \underline{u}, \sigma_\varepsilon^2, D) = K_u \left(\frac{1}{\sigma_\varepsilon^2} \right)^{\frac{1}{2}(q-2)} \left(\frac{1}{\rho} \right) (1-\rho)^{\frac{1}{2}(q-4)} \exp \left\{ -\frac{(1-\rho)}{2\rho\sigma_\varepsilon^2} \underline{u}' A^{-1} \underline{u} \right\} \quad (3.11)$$

and for the variance ratio $v = \frac{\sigma_u^2}{\sigma_\varepsilon^2}$ it follows that

$$p(v | \underline{\beta}, \underline{u}, \sigma_\varepsilon^2, D) = K_u \left(\frac{1}{v\sigma_\varepsilon^2} \right)^{\frac{1}{2}q} \exp \left\{ -\frac{1}{2v\sigma_\varepsilon^2} \underline{u}' A^{-1} \underline{u} \right\}. \quad (3.12)$$

It is clear from equations (3.9) – (3.12) that account of the genetic covariance matrix has been taken in the conditional posterior distributions.

The Gibbs sampling procedure can now easily be implemented. Standard routines are used to generate random numbers from the required distributions. The iterative process starts by using arbitrary starting values $\underline{\beta}^{(0)}$ and $\underline{u}^{(0)}$ in (3.9) and (3.10) to draw $\sigma_\varepsilon^{2(1)}$ and $\sigma_u^{2(1)}$. To complete the first iteration these values are put in (3.7) and (3.8) to draw $\underline{\beta}^{(1)}$ and $\underline{u}^{(1)}$. After k iterations in which the conditional distributions are updated at each iteration the Gibbs sampler has generated the values

$$\underline{\beta}^{(k)}, \underline{u}^{(k)}, \sigma_\varepsilon^{2(k)} \text{ and } \sigma_u^{2(k)}.$$

The process is then repeated m times. The convergence of the algorithm is discussed in Gelfand and Smith (1990). The results for this experiment has been obtained for $k = 20$ and $m = 10000$.

As mentioned, standard routines are used in the simulation process. By using the Matlab package, simulation of $\underline{u}$ for example can be obtained from (3.8) in the following way:

- (i) $\tilde{V}_i \sim N(0, 1) \quad i = 1, \dots, q; \quad \text{i.e.} \quad \tilde{\underline{V}} \sim N(\underline{0}, I_q) \text{ where}$
 $\tilde{\underline{V}} = [\tilde{V}_1 \ \tilde{V}_2, \dots, \tilde{V}_q]'$;
- (ii) $\underline{u} = \sigma_\varepsilon \left(Z'Z + A^{-1} \frac{\sigma_\varepsilon^2}{\sigma_u^2} \right)^{-\frac{1}{2}} \tilde{\underline{V}} + \underline{\tilde{u}}$

where $\underline{\tilde{u}}$ is defined in (3.8). Simulation of $\underline{\beta}$ can be obtained in a similar way.

Simulation of σ_u^2 from (3.10) can be done as follows:

- (i) Simulate τ from a χ_{q-2}^2 -distribution, as a sum of $(q-2)$ squared independent standard normal random deviates, and
- (ii) $\sigma_u^2 = \frac{\underline{u}' A^{-1} \underline{u}}{\tau}.$

Using the posterior conditional densities (2.7) – (2.12) the marginal posterior densities can be obtained by the average of the m conditional densities. These estimates are called Rao-Blackwell estimates (Gelfand and Smith, 1990). For the intraclass correlation coefficient ρ the marginal posterior density illustrated in Section 6 was estimated as

$$\hat{p}(\rho | D) = \frac{1}{m} \sum_{i=1}^m p\left(\rho | \underline{\beta}_i^{(k)}, \underline{u}_i^{(k)}, \sigma_{\varepsilon_i}^{2(k)}, D\right) \quad (3.13)$$

and similarly for the other parameters.

An alternative, perhaps more common form of estimating $\hat{p}(\rho | D)$ is to use samples $\rho_i^{(k)} = \frac{\sigma_{u_i}^{2(k)}}{\sigma_{u_i}^{2(k)} + \sigma_{\varepsilon_i}^{2(k)}} \quad (i = 1, \dots, m)$ only and obtain a kernel density estimator. The Rao-Blackwell estimates are, however, more accurate. Once the unconditional marginal posterior densities are obtained, summary statistics (e.g. the mean and variance) can be computed by a one-dimensional numerical integration procedure, such as Simpson's rule. Probability statements about the unknown parameters can be made as well, thus providing a full solution to inferences about the posterior distributions of the parameters.

4. Review and Extensions of BMOM

In this section the BMOM procedure will be extended to the mixed linear model.

In Section 3 the mixed linear model in its simplest form was defined as

$$\underline{y} = X\underline{\beta} + Z\underline{u} + \underline{\varepsilon} \quad (4.1)$$

In the introductory paragraph it was stated that the BMOM approach is particularly useful where there is difficulty in formulating an appropriate likelihood function. Without a likelihood function, it is not possible to pursue traditional likelihood and Bayesian approaches to estimation and testing.

In this section only the mathematical form of the model as defined in equation (4.1) will be used; i.e. no specific distribution will be assigned to the vector $\underline{\varepsilon}$. The likelihood function will therefore be considered unknown.

This is different from the assumption in Section 2 where it was assumed that $\underline{\varepsilon} \sim N(\underline{0}, \sigma_{\varepsilon}^2 I_n)$; i.e. a known likelihood function.

Let us for the time being assume that $\underline{u}$ is given; (4.1) can then be written as

$$\underline{y} - Z\underline{u} = X\underline{\beta} + \underline{\varepsilon} \quad \text{i.e.}$$

$$\underline{y}^* = X\underline{\beta} + \underline{\varepsilon}. \quad (4.2)$$

For given $\underline{u}$ we will take $\underline{y}^*$ as our new independent variable. Equation (4.2) is now the usual multiple regression model. To assume that the model in (4.2) is adequate implies, among other things, that there are no systematic elements in the realized error vector $\underline{\varepsilon}$, correlated with variables in X . This assumption is formalized as assumption 1 in the BMOM approach as follows:

Assumption 1

$$X'E(\underline{\varepsilon} | D, \underline{u}) = \underline{0}$$

where $E(\underline{\varepsilon} | D, \underline{u})$ denotes the post-data mean of $\underline{\varepsilon}$ given the data and $\underline{u}$. That is, the given, unknown values of the elements of the realized error vector $\underline{\varepsilon}$ are considered subjectively random, just as in Bayesian analysis of realized error terms. See for example, Chaloner and Brant (1988), Chaloner (1994), Zellner and Moulton (1985) and Zellner (1975). Thus Assumption 1 indicates that the columns of X are orthogonal to the vector $E(\underline{\varepsilon} | D, \underline{u})$. Further from (4.2), by taking the posterior expectation, it follows that

$$\underline{y}^* = XE(\underline{\beta} | D, \underline{u}) + E(\underline{\varepsilon} | D, \underline{u})$$

and Assumption 1 implies that the observation vector $\underline{y}$ is the sum of two orthogonal vectors. Note also since the first column of X is an $n \times 1$ vector of ones, denoted by $\underline{1}$, we have from Assumption 1, $\underline{1}'E(\underline{\varepsilon} | D, \underline{u}) = 0$, or $E\left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i\right) = E(\bar{\varepsilon} | D, \underline{u}) = 0$. Thus given that we assume that we have an appropriate form and an adequate number of terms included in (4.2), the expectation of the mean of the realized error terms is assumed equal to zero; that is there is no systematic component in the realized error vector.

By multiplying both sides of (4.2) by $(X'X)^{-1}X'$ we have (see for example Zellner, 1997a, Tobias and Zellner, 1997, and Zellner *et al.*, 1999) that

$$\underline{\hat{\beta}} = (X'X)^{-1}X'\underline{y}^* = \underline{\beta} + (X'X)^{-1}X'\underline{\varepsilon}. \quad (4.3)$$

On taking post-data expectations of both sides of (4.3) given the data $D = (\underline{y}, X)$ and $\underline{u}$ it follows that

$$\underline{\hat{\beta}} = E(\underline{\beta} | D, \underline{u}) + (X'X)^{-1}X'E(\underline{\varepsilon} | D, \underline{u}). \quad (4.4)$$

and from Assumption 1 that

$$E(\underline{\beta} | D, \underline{u}) = \underline{\hat{\beta}} = (X'X)^{-1}X'\underline{y}^* \quad (4.5)$$

That is the post-data expectation of the regression coefficient vector is equal to the least squares estimate. Further the post data mean of the realized error vector $\underline{\varepsilon}$ in (4.2) is

$$E(\underline{\varepsilon} | D, \underline{u}) = \underline{y}^* - X\underline{\hat{\beta}} = \underline{\hat{\varepsilon}} \quad (4.6)$$

where $\underline{\hat{\varepsilon}}$ is the least squares residual vector that satisfies $X'\underline{\hat{\varepsilon}} = \underline{0}$. Note also that from (4.3) and (4.6) we can write

$$\begin{aligned} \underline{\varepsilon} - \underline{\hat{\varepsilon}} &= \underline{y}^* - X\underline{\beta} - (\underline{y}^* - X\underline{\hat{\beta}}) \\ &= \underline{y}^* - X\underline{\beta} - \left\{ \underline{y}^* - X \left(\underline{\beta} + (X'X)^{-1}X'\underline{\varepsilon} \right) \right\} \end{aligned}$$

$$\begin{aligned}
&= X (X'X)^{-1} X' \underline{\varepsilon} \\
&= X (X'X)^{-1} X' (\underline{\varepsilon} - \hat{\underline{\varepsilon}})
\end{aligned} \tag{4.7}$$

where the last step follows from the orthogonality condition mentioned above, $X' \hat{\underline{\varepsilon}} = \underline{0}$.

We can thus write

$$\begin{aligned}
Var(\underline{\varepsilon} | D, \underline{u}) &= E \{ (\underline{\varepsilon} - \hat{\underline{\varepsilon}}) (\underline{\varepsilon} - \hat{\underline{\varepsilon}})' | D, \underline{u} \} \\
&= X (X'X)^{-1} X' E \{ (\underline{\varepsilon} - \hat{\underline{\varepsilon}}) (\underline{\varepsilon} - \hat{\underline{\varepsilon}})' | D, \underline{u} \} X (X'X)^{-1} X'
\end{aligned} \tag{4.8}$$

which defines a functional equation that the post data covariance matrix for $\underline{\varepsilon}$, $Var(\underline{\varepsilon} | D, \underline{u})$ must satisfy. Since there are only p free elements of $\underline{\varepsilon}$ in the n equations in (4.2), $Var(\underline{\varepsilon} | D, \underline{u})$ must be of rank p . Thus we introduce the following assumption that fixes the form of the realized error vector covariance matrix up to a multiplicative positive scalar multiplier.

Assumption 2

$$Var(\underline{\varepsilon} | \sigma_\varepsilon^2, D, \underline{u}) = \sigma_\varepsilon^2 X (X'X)^{-1} X'$$

where σ_ε^2 is a variance parameter to be defined below. We use Assumption 2 to evaluate the post-data covariance matrix of $\underline{\beta}$ as follows:

$$\begin{aligned}
Var(\underline{\beta} | \sigma_\varepsilon^2, D, \underline{u}) &= E \left[(\underline{\beta} - \hat{\underline{\beta}}) (\underline{\beta} - \hat{\underline{\beta}})' | D, \underline{u} \right] \\
&= (X'X)^{-1} X' E \left[(\underline{\beta} - \hat{\underline{\beta}}) (\underline{\beta} - \hat{\underline{\beta}})' | D, \underline{u} \right] X (X'X)^{-1} \\
&= \sigma_\varepsilon^2 (X'X)^{-1}
\end{aligned} \tag{4.9}$$

where $\sigma_\varepsilon^2 = \frac{1}{n} \sum_{i=1}^n \varepsilon_i^2 = \frac{1}{n} \underline{\varepsilon}' \underline{\varepsilon}$. It is seen that the parameter σ_ε^2 is represented as an average of the sum of squared deviations of the realized errors from their expected mean of zero, $E(\underline{\varepsilon} | D, \underline{u}) = 0$ which follows from Assumption 1.

Using this definition of σ_ε^2 we have

$$\begin{aligned} E(\sigma_\varepsilon^2 | D, \underline{u}) &= E \frac{1}{n} (\underline{\varepsilon}' \underline{\varepsilon} | D, \underline{u}) \\ &= E \frac{1}{n} \left\{ (\underline{y}^* - X \underline{\beta})' (\underline{y}^* - X \underline{\beta}) | D, \underline{u} \right\} \\ &= E \frac{1}{n} \left[\left\{ (\underline{y}^* - X \underline{\hat{\beta}}) - (X \underline{\beta} - X \underline{\hat{\beta}}) \right\}' \left\{ (\underline{y}^* - X \underline{\hat{\beta}}) - (X \underline{\beta} - X \underline{\hat{\beta}}) \right\} | D, \underline{u} \right] \\ &= E \frac{1}{n} \left[\left\{ (\underline{y}^* - X \underline{\hat{\beta}})' (\underline{y}^* - X \underline{\hat{\beta}}) - 2 (\underline{y}^* - X \underline{\hat{\beta}})' (X \underline{\beta} - X \underline{\hat{\beta}}) \right. \right. \\ &\quad \left. \left. + (\underline{\beta} - \underline{\hat{\beta}})' (X' X) (\underline{\beta} - \underline{\hat{\beta}}) | D, \underline{u} \right\} \right]. \end{aligned}$$

Since the middle term is equal to zero ($\underline{y}^* X - \underline{\hat{\beta}}' X' X = 0$), it follows that

$$\begin{aligned} E(\sigma_\varepsilon^2 | D, \underline{u}) &= \frac{1}{n} \left[\underline{\hat{\varepsilon}}' \underline{\hat{\varepsilon}} + E \left\{ (\underline{\beta} - \underline{\hat{\beta}})' (X' X) (\underline{\beta} - \underline{\hat{\beta}}) | D, \underline{u} \right\} \right] \\ &= \frac{1}{n} \{ \underline{\hat{\varepsilon}}' \underline{\hat{\varepsilon}} + p E(\sigma^2 | D, \underline{u}) \} \end{aligned}$$

$$\text{or } E(\sigma_\varepsilon^2 | D, \underline{u}) = \frac{\underline{\hat{\varepsilon}}' \underline{\hat{\varepsilon}}}{n - p} = s^2. \quad (4.10)$$

Note that this post-data expectation differs from the post-data mean of σ_ε^2 in a diffuse prior-normal likelihood traditional Bayesian approach, namely

$E_{TB}(\sigma_\varepsilon^2 | D, \underline{u}) = \frac{\nu s^2}{(\nu - 2)}$. For small values of $\nu = n - p$, the last expression is much larger than s^2 .

As pointed out in Zellner (1996), Tobias and Zellner (1997) and mentioned in Section 1, the proper maxent density for $\underline{\beta}$ given σ_ε^2 , D and $\underline{u}$ using the first two moments is the normal distribution; i.e.

$$f_N(\underline{\beta} | \sigma_\varepsilon^2, D, \underline{u}) \sim N[\hat{\underline{\beta}}, (X'X)^{-1} \sigma_\varepsilon^2] \quad (4.11)$$

and using the first moment of σ_ε^2 given in (4.10) the proper maxent density for σ_ε^2 is an exponential density,

$$h_\varepsilon(\sigma_\varepsilon^2 | D, \underline{u}) = (1/s^2) \exp\{-\sigma_\varepsilon^2 / s^2\} \quad 0 < \sigma_\varepsilon^2 < \infty \quad (4.12)$$

which will be called BMOM 1.

From (4.11) and (4.12) it follows that

$$p(\underline{\beta}, \sigma_\varepsilon^2 | D, \underline{u}) = f_N(\underline{\beta} | \sigma_\varepsilon^2, D, \underline{u}) h_\varepsilon(\sigma_\varepsilon^2 | D, \underline{u}) \quad (4.13)$$

Also as shown in Tobias and Zellner (1997) and Zellner (1997a), higher order post-data moments of σ_ε^2 can be evaluated and used as moment side conditions in deriving maxent densities. Pearson curve approximations can also be used if the first four moments are available. This approach will be called BMOM2.

These higher order post data moments of σ_ε^2 can be obtained in the following way. From the definition of σ_ε^2 , it follows that

$$\begin{aligned} \sigma_\varepsilon^2 &= \frac{1}{n} \underline{\varepsilon}' \underline{\varepsilon} = \frac{1}{n} \left(\nu s^2 + (\underline{\beta} - \hat{\underline{\beta}})' (X'X) (\underline{\beta} - \hat{\underline{\beta}}) \right) \\ &= \frac{1}{n} (\nu s^2 + \sigma_\varepsilon^2 Q) \end{aligned} \quad (4.14)$$

and from equation (4.11) it follows that $Q = \frac{1}{\sigma_\varepsilon^2} (\underline{\beta} - \hat{\underline{\beta}})' (X'X) (\underline{\beta} - \hat{\underline{\beta}})$ has a chi-square density with p degrees of freedom. As shown by Tobias and

Zellner (1997) this fact can be employed to evaluate the moments of σ_ε^2 as illustrated below.

From (4.14) it follows that

$$\sigma_\varepsilon^{2j} = \frac{1}{n^j} (\nu s^2 + \sigma_\varepsilon^2 Q)^j; \quad j = 1, 2, \dots \quad (4.15)$$

By using the binomial expansion and known moments of χ_p^2 the following recursive formulae can be derived:

$$E(\sigma_\varepsilon^{2j} | D, \underline{u}) = \frac{(\nu s^2)^j + \sum_{i=1}^{j-1} \binom{j}{i} (\nu s^2)^{j-i} E(\sigma_\varepsilon^{2i} | D) [p(p+2) \dots (p+2(i-1))]}{n^j - [p(p+2) \dots (p+2(j-1))]} \quad (4.16)$$

From this expression the first two moments above zero and the variance of σ_ε^2 is given by

$$E(\sigma_\varepsilon^2 | D, \underline{u}) = s^2, \quad E(\sigma_\varepsilon^4 | D, \underline{u}) = \frac{s^4 (\nu^2 + 2\nu p)}{n^2 - p(p+2)}$$

$$\text{and } Var(\sigma_\varepsilon^2 | D, \underline{u}) = s^4 \left(\frac{2p}{n^2 - p(p+2)} \right). \quad (4.17)$$

A similar approach as just described can be used to obtain the post data densities of $\underline{u}$ and σ_u^2 .

For the mixed linear model defined in (4.1) we can also assume that $\underline{\beta}$ instead of $\underline{u}$ is known.

Equation (4.1) can then be written as

$$\underline{y} - X\underline{\beta} = Z\underline{u} + \varepsilon, \quad \text{i.e.}$$

$$\tilde{\underline{y}} = Z\underline{u} + \underline{\varepsilon}. \quad (4.18)$$

Using the same arguments as given in equations (4.3) – (4.11) it follows that the maxent density for $\underline{u}$ is normal with mean, $\hat{\underline{u}} = (Z'Z)^{-1} Z'\tilde{\underline{y}}$ and variance $\hat{\sigma}_u^2 = (Z'Z)^{-1} \sigma_\varepsilon^2$. To implement the normal prior for $\underline{u}$ (which is an integral

part of the mixed linear model analyses) the likelihood of $\underline{u}$ given σ_ε^2 and $\underline{\beta}$ is considered to be proportional to the maxent density. Multiplying the likelihood with the prior (equation (3.4)) gives

$$e^{-\frac{1}{2}(\underline{u}-\underline{\hat{u}})'(Z'Z)(\underline{u}-\underline{\hat{u}})/\sigma_\varepsilon^2} \times e^{-\frac{1}{2}\underline{u}'A^{-1}\underline{u}/\sigma_u^2}$$

which implies that the posterior distribution of $\underline{u}$ is normal with density

$$P_N(\underline{u} | \sigma_\varepsilon^2, \sigma_u^2, D, \underline{\beta}) \sim N_q \left[\underline{\tilde{u}}, \left(Z'Z + A^{-1} \frac{\sigma_\varepsilon^2}{\sigma_u^2} \right)^{-1} \sigma_\varepsilon^2 \right]$$

$$\text{where } \underline{\tilde{u}} = \left(Z'Z + A^{-1} \frac{\sigma_\varepsilon^2}{\sigma_u^2} \right)^{-1} Z'(\underline{u} - X\underline{\beta}). \quad (4.19)$$

Equation (4.19) is identical to the conditional posterior derived by the traditional Bayesian method (equation (3.8)). Also the conditional posterior distribution of σ_u^2 in the BMOM case is identical to (3.10), this follows from the normal prior density (equation (3.4)).

The Gibbs sampler to generate random numbers from the required distributions will therefore be implemented in the same way as described for the traditional Bayesian approach with one exception that in the BMOM case σ_ε^2 will be sampled from (4.12) (the exponential density, BMOM 1) or from any other maxent density of σ_ε^2 . In the example that follows the first four moments of σ_ε^2 will be calculated using the recursive equation (4.16) and a Pearson curve approximation will be used for the conditional posterior distribution of σ_ε^2 .

5. Another Bayesian Method of Moments Approach for the Mixed Linear Model

In this section another approach to the BMOM analysis will be given which is "more distribution free" or "less likelihood" than the previous one.

As in Section 4 the proper maxent densities of $\underline{\beta}$ and σ_ε^2 will be obtained by using Assumptions 1 and 2. These densities are given in equations (4.11) and (4.12). Higher order post data moments of σ_ε^2 can also be calculated and used as moment side conditions in deriving other maxent densities.

The derivations that follow will, however, be different from those given in equations (4.18) and (4.19).

Substitute starting values $\underline{u}^{(0)}$ and $\underline{\beta}^{(0)}$ in equation (4.1), to calculate

$$\underline{\varepsilon}^{(0)} = \underline{y} - X\underline{\beta}^{(0)} - Z\underline{u}^{(0)}. \quad (5.1)$$

Also for given $\underline{u}^{(0)}$, draw $\sigma_\varepsilon^{2(1)}$ and $\underline{\beta}^{(1)}$ from (4.12) and (4.11). A new $\underline{u}$ which will be called

$$\underline{u}_{\text{post}}^{(1)} = (Z'Z)^{-1} Z' \left(\underline{y} - X\underline{\beta}^{(1)} - \underline{\varepsilon}^{(0)} \right) \quad (5.2)$$

can now be calculated as well as

$$\sigma_u^{2(1)} = \frac{1}{q} \left(\underline{u}_{\text{post}}^{(1)} A^{-1} \underline{u}_{\text{post}}^{(1)} \right) \quad (5.3)$$

where "post" means "posterior".

To implement the normal prior assumption for $\underline{u}$ (equation (3.4)) draw a $\underline{u}_{\text{prior}}^{(1)}$ from the normal distribution $N(\underline{0}, \sigma_u^{2(1)} A)$ and calculate

$$\underline{\varepsilon}^{(1)} = \underline{y} - X\underline{\beta}^{(1)} - Z\underline{u}_{\text{prior}}^{(1)} \quad (5.4)$$

to complete the first iteration. After k iterations in which the conditional distributions were updated at each iteration, the Gibbs-sampler has generated the values $\underline{\beta}^{(k)}$, $\underline{u}_{\text{post}}^{(k)}$, $\sigma_\varepsilon^{2(k)}$ and $\sigma_u^{2(k)}$. The process is then repeated m times

For our practical problem the BMOM posterior densities obtained using this section and those derived from the previous section were for all practical purposes the same.

It is therefore clear that in the case of the BMOM analysis the posterior moments, based on a few relatively weak assumptions, can be used to obtain post data densities (maximum entropy densities) for parameters and realized error terms without the use of the likelihood function or prior density.

The assumption of no prior information has as consequence different types of derivations (post data densities) that differ from those obtained using the traditional Bayesian approach where prior information was assigned to the unknown parameters $\underline{\beta}$, σ_ε^2 , $\underline{u}$ and σ_u^2 .

As mentioned, no prior densities are necessary for the BMOM procedure, but if some prior information is available it can be built into the BMOM procedure. Since the assumption of a normal prior for the random effects $\underline{u}$ is an integral part of the mixed linear model, the prior density $\underline{u} \sim N(\underline{0}, \sigma_u^2 A)$ was also used in the BMOM analysis.

6. Results and Discussions

6.1 Bayes, REML and BMOM Estimates of the Variance Components

The estimates of the variance components for the sire model, are presented in Table 1. The estimate of σ_ε^2 given in the first row and first column of the table is the Bayes estimate obtained from the traditional Bayes approach.

The following two estimates of σ_ε^2 given in Table 1 are the estimates obtained from the BMOM procedure, drawing first from an exponential distribution (BMOM 1) and then from a Pearson Type 4 curve (BMOM 2).

The corresponding marginal posterior densities of σ_ϵ^2 are presented in Figures 1, 2 and 3. The fourth estimate of σ_ϵ^2 is the Restricted Maximum Likelihood (REML) estimate.

The corresponding estimates of the sire variance (σ_u^2) (Modes of the posterior distribution) are given in the second column, namely 3.01 (Traditional Bayes estimate), 3.86 (BMOM 1) and 3.16 (BMOM 2). These estimates do not differ much from the REML estimate which is 3.08. The posterior densities of σ_u^2 are quite skewed (Figure 4), this means that the means, modes and medians will not coincide.

Table 1

	σ_ϵ^2	σ_u^2	95% Credibility Interval for σ_ϵ^2	95% Credibility Interval for σ_u^2
Traditional Bayes	21.2595	3.01	19.3130 – 23.2531	1.2461 – 12.1343
BMOM 1	21.1125	3.86	0.0000 – 64.5875	1.5651 – 14.3427
BMOM 2	20.2873	3.16	19.9979 – 20.6214	1.2464 – 13.1652
REML	21.1096	3.08	–	–

From Table 1 it is clear that the posterior mean of σ_ϵ^2 is 21.1125 in the case of BMOM 1; the posterior median on the other hand is 18.79 while the mode is virtually zero. The posterior density displayed in Figure 1 is quite skewed and that is the reason for the large discrepancies in central values.

6.2 Density estimation of functions of variance components

Using the posterior conditional densities for σ_ϵ^2 and σ_u^2 Gibbs sampling, the marginal posterior densities are displayed in Figures 1 – 4.

Using the theory of transformation of random variables as described in Section 3, we can also obtain the conditional posterior density of the intraclass correlation $\rho = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2}$ and variance ratio coefficient $v = \frac{\sigma_u^2}{\sigma_\varepsilon^2}$, by fixing σ_ε^2 .

The marginal posterior densities of ρ and v are illustrated in Figures 5 and 6. The estimated values for the intraclass correlation as well as a 95% credibility intervals are given in Table 2.

Table 2.
95% Credibility Intervals and Intraclass Correlation Coefficients
obtained from the Gibbs Sampler

	ρ	95% Interval
Traditional Bayes	0.1789	0.0550–0.3658
BMOM 1	0.3303	0.0417–0.8351
BMOM 2	0.1886	0.0577–0.3944

Except for BMOM 1, the 95% credibility interval ρ for the intraclass correlation does not contain 0.5. This result corresponds to the statement made by Wang *et al.* (1993), namely that from a genetic point of view, an intraclass correlation of 0.5 is not possible in a Sire model. Posterior densities of other functions of the variance components can be obtained in a similar manner.

The 95% credibility interval for ρ in the case of BMOM 1 differs substantially from the corresponding intervals for BMOM 2 and Traditional Bayes. This was as expected because the exponential density (the maxent using only one moment) is quite skew. In practice usually two or more moments are available. The credibility intervals for BMOM 2 and Traditional Bayes on the other hand are for all practical purposes the same.

We can therefore conclude that the traditional Bayesian and BMOM approaches to variance component estimation have several practical advantages over a classical approach. Firstly, although the estimate for a variance component is always positive, the REML estimate's asymptotic distribution can generate interval estimates that include negative values. An interval estimate such as a highest posterior density region will not include negative values. This potentially embarrassing phenomenon is often overlooked in the discussions of likelihood-based methods. Secondly, highest posterior density regions are never empty, whereas confidence intervals for the ratio of two variances can be empty. One can also report the whole of the posterior probability distribution, not just a single number, and report some measure of posterior precision. Finally, classical estimators generally have intractable sampling distributions and standard errors are hard to calculate (Van der Merwe and Botha, 1993).

6.3 Density Estimation for Random and Fixed Effects

Posterior distributions of the random effects can be obtained directly from the Gibbs sampler. The estimated breeding values for 17 sires are given in Table 3. The REML estimates and their corresponding standard errors are given in the last two columns.

It is evident from Table 3 that the estimates using the different procedures are quite close to one another. The Traditional Bayes and the BMOM 2 estimates are for all practical purposes the same. Also, it can be seen that the estimated breeding values (mean values for BMOM 2) for sires 3 and 10 are 3.4931 and -1.8861 respectively. Progeny from sire 3 with an estimated breeding value of 3.4931 will therefore have an estimated average weaning weight of 3.4931 kilogram more than the progeny from the rest of the sires. The progeny from sire 10 on the other hand with an estimated breeding value of -1.8861 will have

an estimated average weaning weight of 1.8861 kilogram less than the average weaning weight of lambs from the rest of the sires. The marginal posterior densities of these two sires are presented in Figures 7 and 8. The estimated posterior density of the difference in breeding values between sires u_3 and u_{10} is presented in Figure 9.

Table 3

Estimated Breeding Values for 17 Sires from the Elsenburg Dormer Stud (1943–1950) as well as SEs of REML Estimates

Sire	no.	Trad. Bayes	BMOM 1	BMOM 2	REML	SEs REML
41037	1	0.7350	0.8889	0.7098	0.5781	1.06
41004	2	0.2478	0.3397	0.2415	0.1396	0.92
41019	3	3.4858	3.6370	3.4931	3.3300	0.99
43002	4	-1.1985	-1.3089	-1.2460	-1.1810	1.18
44170	5	-0.0930	-0.0387	-0.0943	-0.1700	1.18
44174	6	-0.6524	-0.7942	-0.6847	-0.5694	1.34
44042	7	-1.3053	-1.3338	-1.3560	-1.2565	0.95
45070	8	-1.1460	-1.1793	-1.1833	-0.9631	0.93
45135	9	-0.5301	-0.5069	-0.5984	-0.5371	1.10
46015	10	-1.7983	-1.8758	-1.8861	-1.7092	0.96
46037	11	-0.8524	-0.8960	-0.9098	-0.8423	0.91
48014	12	-1.0059	-1.1001	-1.0759	-0.9537	0.97
48052	13	-0.4208	-0.4541	-0.4708	-0.3019	1.00
48140	14	-1.4307	-1.5019	-1.4750	-1.2560	1.10
49053	15	0.5309	0.4535	0.4479	0.4630	1.31
49134	16	0.9219	0.7731	0.9641	0.7950	1.34
49046	17	0.4395	0.3892	0.3379	0.4059	1.41

Posterior distributions of the fixed effects can also be obtained directly from the Gibbs sampler. The estimated values of β_7 , β_{14} , β_{15} and β_{16} are given in tables 4, 5 and 6.

Table 4
Estimated Values of Selected Fixed Effects as well as 95% Credibility Intervals for the Traditional Bayesian Method

Fixed Effect	Estimated Value	95% Credibility Interval
β_7	5.3523	4.1515 – 6.5310
β_{14}	3.6690	2.9835 – 4.3353
β_{15}	9.4874	7.1923 – 11.7688
β_{16}	2.9621	0.6574 – 5.2308

Table 5
Estimated Values of Selected Fixed Effects as well as 95% Credibility Intervals for BMOM 1

Fixed Effect	Estimated Value	95% Credibility Interval
β_7	5.2995	4.1210 – 6.3710
β_{14}	3.6757	3.0873 – 4.3134
β_{15}	9.5505	7.5584 – 11.6362
β_{16}	3.0469	1.0836 – 5.0166

Table 6
Estimated Values of Selected Fixed Effects as well as 95% Credibility Intervals for BMOM 2

Fixed Effect	Estimated Value	95% Credibility Interval
β_7	5.3037	4.1175 – 6.4561
β_{14}	3.6854	3.0037 – 4.2923
β_{15}	9.4580	7.2886 – 11.5086
β_{16}	2.9574	0.7831 – 5.0853

The estimated marginal posterior densities of these four fixed effects are presented in Figures 10 – 13. β_7 measures the expected difference in average

weaning weight between lambs born in 1949 and lambs born in 1950. It is therefore clear that lambs born in 1949 had an average weaning weight of 5.3037 (BMOM 2) kilogram more than lambs born in 1950. The 95% credibility interval is 4.1175 – 6.4561.

β_{14} measures the expected difference in average weaning weight between male and female lambs. It can therefore be concluded that male lambs will have an average weaning weight of 3.6854 kilogram more than female lambs. From the 95% credibility interval it can be seen that the difference in average weaning weight between male and female lambs can be as large as 4.2923 kilogram.

β_{15} measures the expected difference in average weaning weight between single births and triplets. It can therefore be expected that single births will have an average weaning weight of 9.4580 kilogram more than triplets.

β_{16} measures the expected difference in average weaning weight between twins at birth and triplets. Therefore, twins at birth will have an average weaning weight of 2.9574 kilogram more than triplets. Therefore, the birth status can dramatically affect the expected weaning weight of a sire's progeny, thus affecting its breeding value. According to van Wyk (1992) single born lambs constitute only 36.02% of all lambs born while twins and triplets made up 59.93% and 4.05% respectively.

Posterior densities of linear combinations of fixed effects can also be derived using the Gibbs sampler.

7. Conclusion

It was mentioned by one of the referees that the example and its treatment are rather disappointing because the final comparison shows virtually no difference between Bayesian and non-Bayesian results.

This is not surprising to us, since as showed by Harville, D.A. (1974), Bayesian Inference for Variance Components using only Error Contrasts. *Biometrika*, 61: 383–385 (see also Searle, S.R., Casella, G. and McCulloch, C.E., 1992, *Variance Components*, John Wiley and Sons, Inc. New York) that when “flat” or uniform priors are assigned to the vector of fixed effects and the variance components the modes of the marginal posterior distributions are very close to the REML estimates.

If on the other hand proper priors were assigned to the unknown parameters and if the sample size was quite small, the differences between the Bayesian and non-Bayesian results could have been quite substantial. The assignment of a proper prior to a specific parameter must however be justifiable from a practical point of view. In some animal breeding experiments for example it is known beforehand that $\sigma_u^2 < \frac{1}{3}\sigma_\varepsilon^2$. This information can if necessary be used to formulate a proper prior on the interval $\left[0, \frac{1}{3}\right]$ for the variance ratio $\gamma = \frac{\sigma_u^2}{\sigma_\varepsilon^2}$. The Gibbs sampler can easily be adjusted to handle this case. The assignment of proper priors was, however, not the purpose of this note.

Inspection of Figures 4, 5, 6, 10, 11 and 12 shows that the differences between the traditional Bayesian and BMOM results can be quite substantial (especially in the case of BMOM1). As mentioned for the BMOM analysis no likelihood was assumed. The post data densities therefore depend on the form of the maximum entropy densities which in turn depend on the number of moments used. In the case of BMOM1 the posterior conditional density of σ_ε^2 for given $\underline{u}$ is the exponential density (maxent density using only the first moment). The double exponential effect of the marginal posterior densities of the $\underline{\beta}$ parameters in the case of BMOM1 can easily be recognised from Figures 10 11 and 12.

The differences between the traditional Bayes and BMOM results (as displayed in the figures) could have been much larger, if as mentioned, proper priors were assigned to the unknown parameters in the traditional Bayesian analysis.

It can therefore be concluded that the BMOM approach yields useful and predictive inferences without using assumed likelihood functions, prior densities for the parameters and Bayes' theorem. On introducing simple assumptions relating to the moments of the realized error terms and the future, as yet unobserved error terms, we could derive post data moments and future values of the dependent variables. Thus, in summary, it is the case that BMOM techniques provide solutions to inverse problems in situations in which there are difficulties in applying traditional likelihood or Bayesian analyses.

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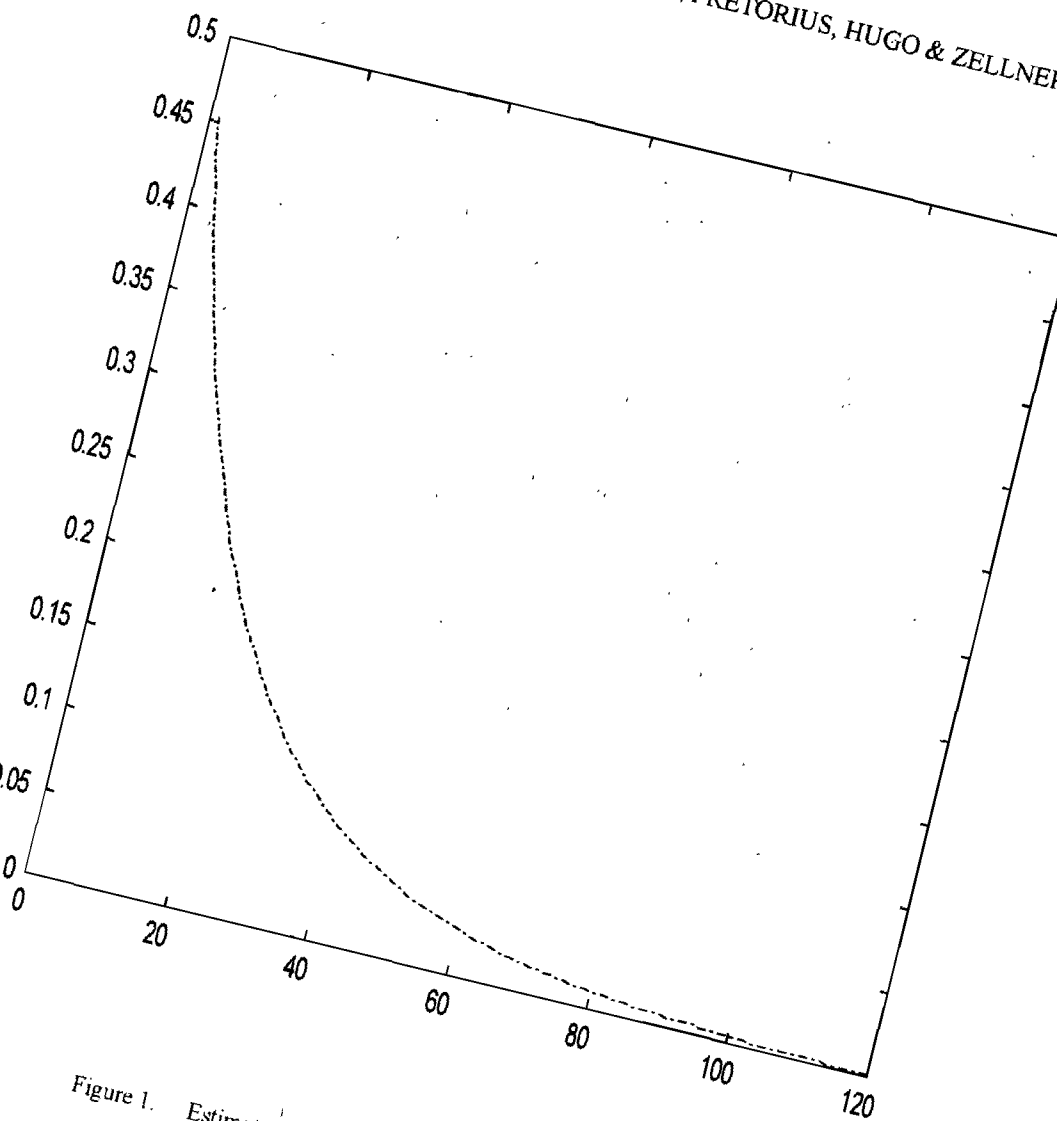


Figure 1. Estimated posterior density of the error variance, σ_{ϵ}^2 for BMOM 1

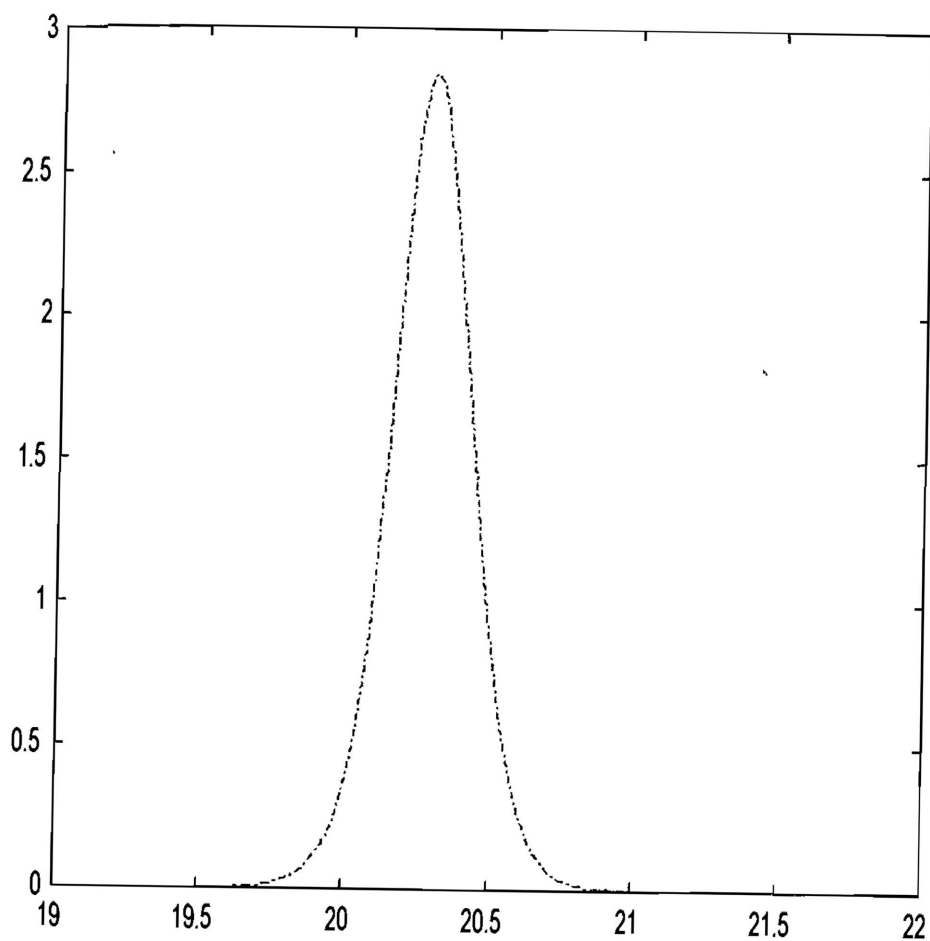


Figure 2. Estimated posterior density of the error variance, σ_{ϵ}^2 for BMOM 2

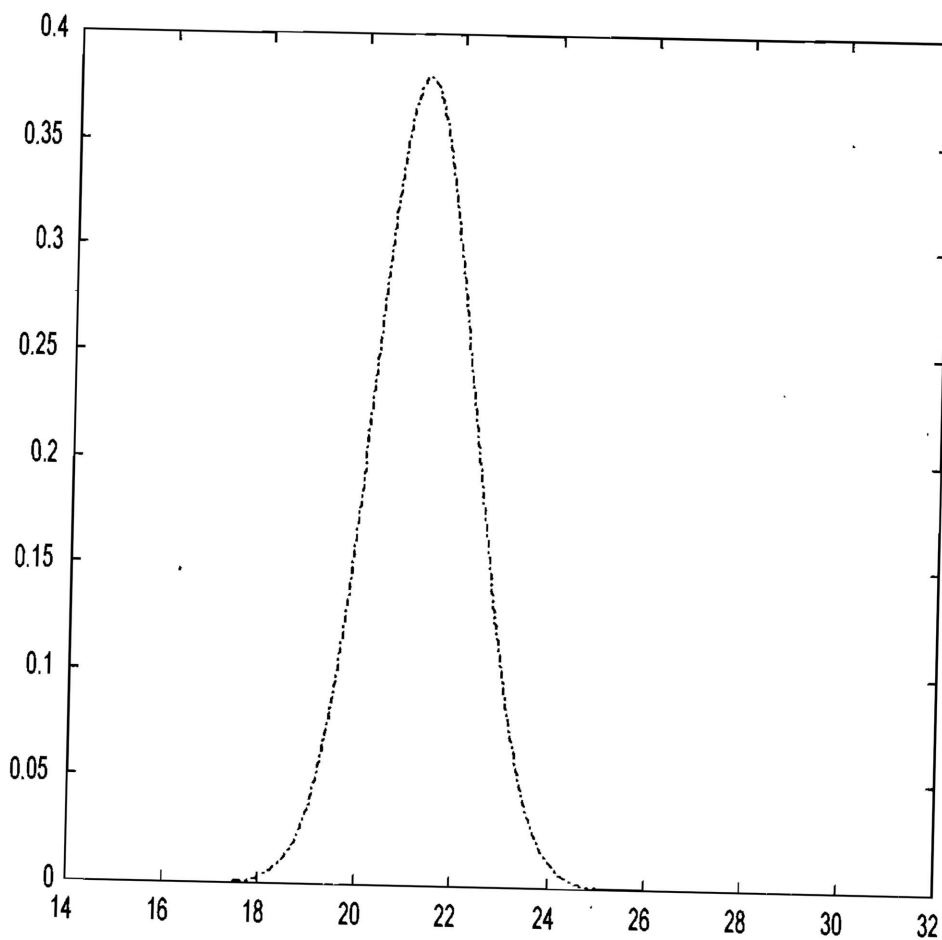


Figure 3. Estimated posterior density of the error variance, σ_{ε}^2 for Traditional Bayes

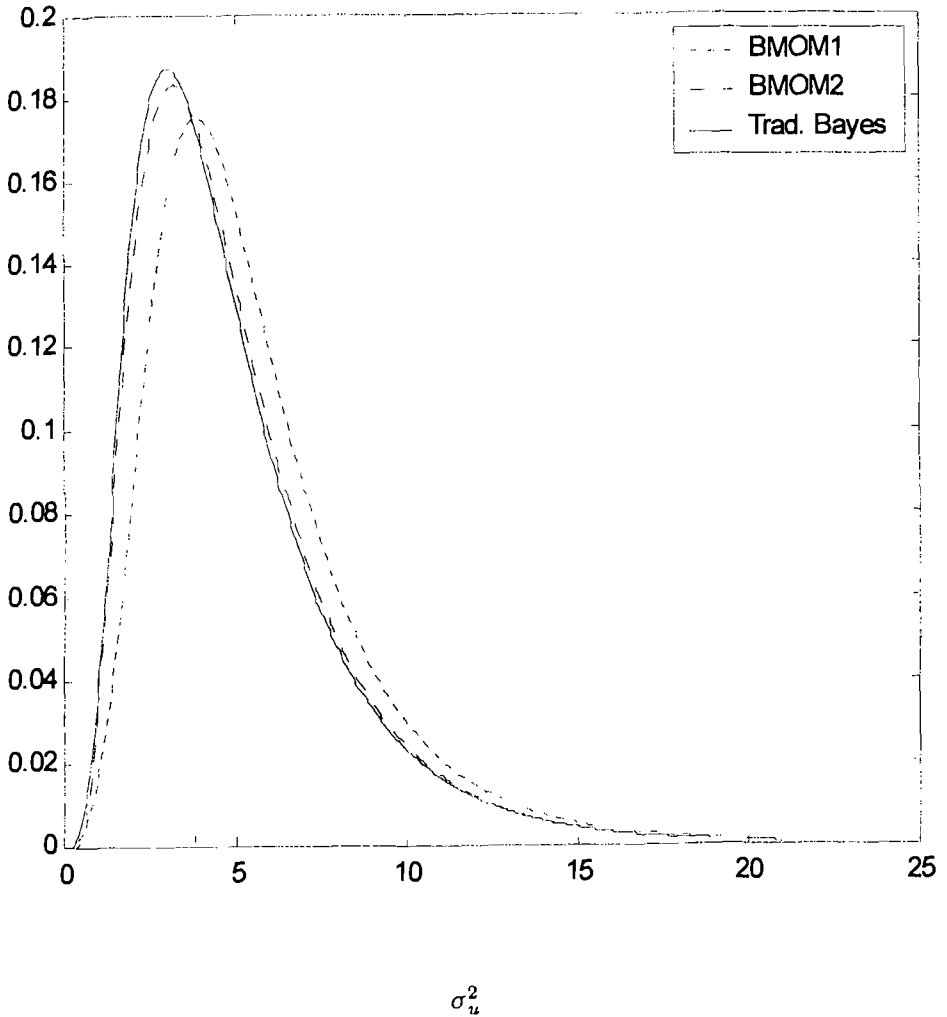


Figure 4. Estimated posterior density of the sire variance, σ_u^2
BMOM 1 (Mean = 5.6778; Mode = 3.86),
BMOM 2 (Mean = 5.0361; Mode = 3.16);
Trad. Bayes (Mean = 4.9239; Mode = 3.01).

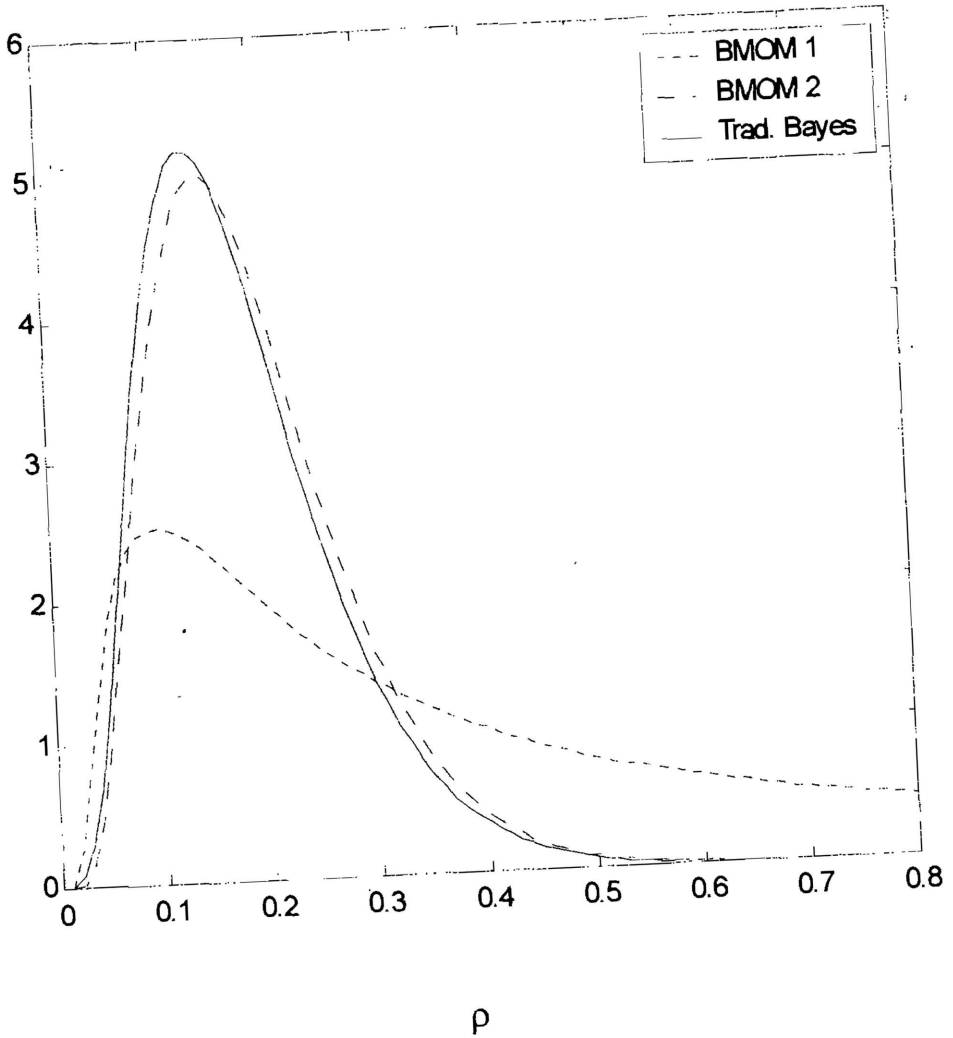


Figure 5. Estimated posterior density of the Intraclass Correlation Coefficient

BMOM (1)	Mean = 0.33	Mode = 0.10
BMOM (2)	Mean = 0.19	Mode = 0.15
Traditional Bayes	Mean = 0.18	Mode = 0.14

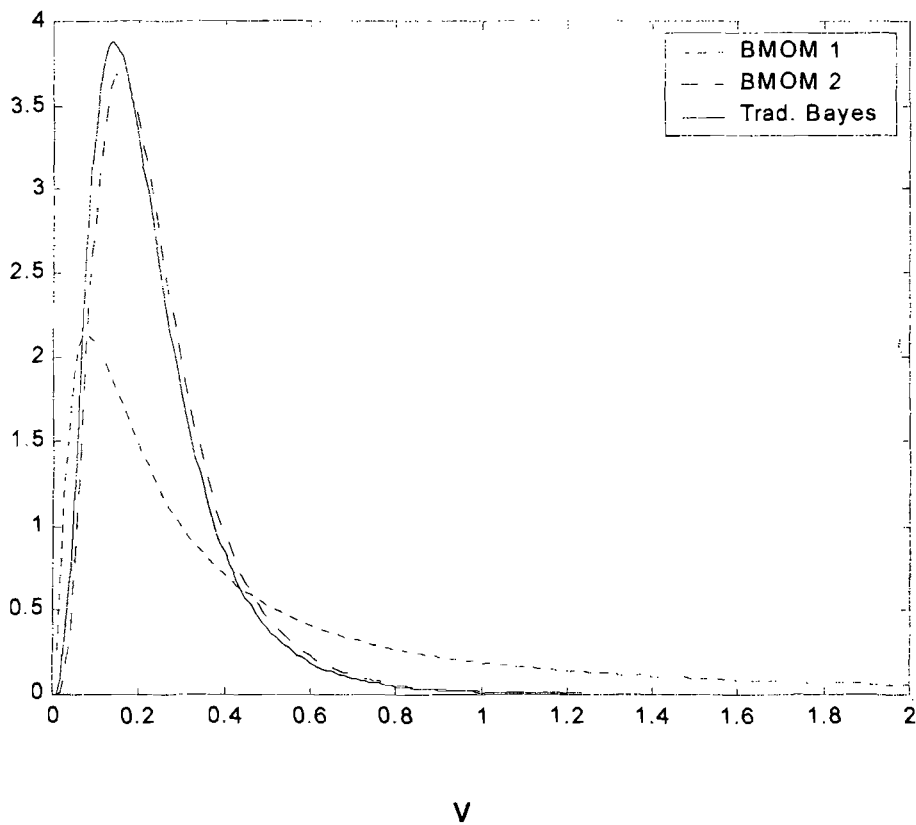


Figure 6. Estimated posterior density of the variance ratio coefficient, v

BMOM (1)	Mean =	3.2125	Mode =	0.08
BMOM (2)	Mean =	0.2486	Mode =	0.16
Traditional Bayes	Mean =	0.2326	Mode =	0.14

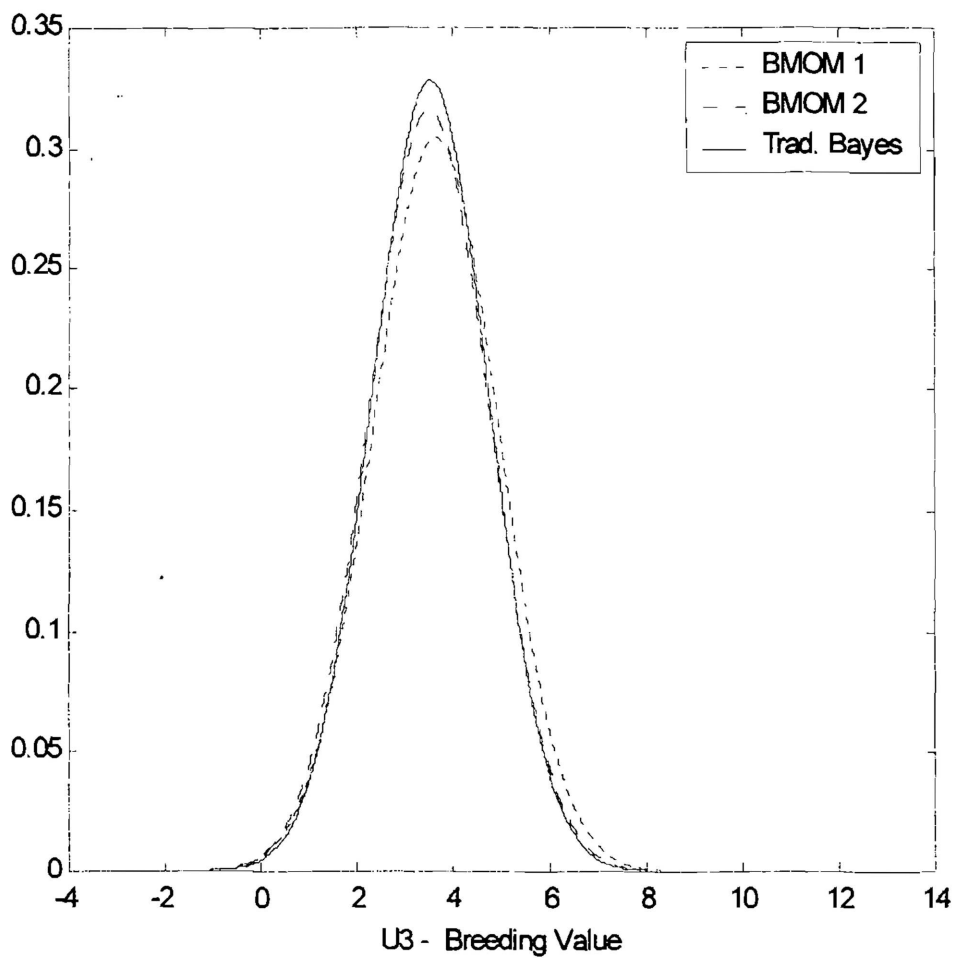


Figure 7. Estimated posterior density of the breeding value for Sire 3
BMOM 1 (Mean = 3.64), BMOM 2 (Mean = 3.49),
Trad. Bayes (Mean = 3.49)

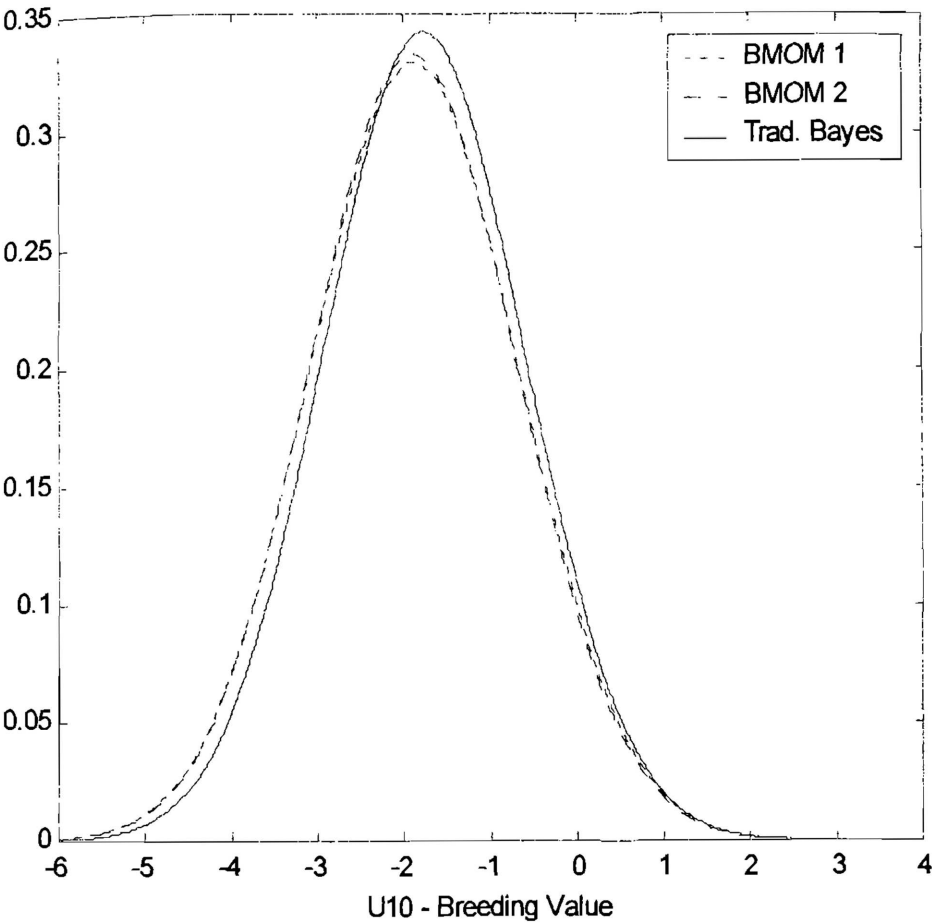


Figure 8. Estimated posterior density of the breeding value for Sire 10
BMOM 1 (Mean = -1.88), BMOM 2 (Mean = -1.88),
Trad. Bayes (Mean = -1.80)

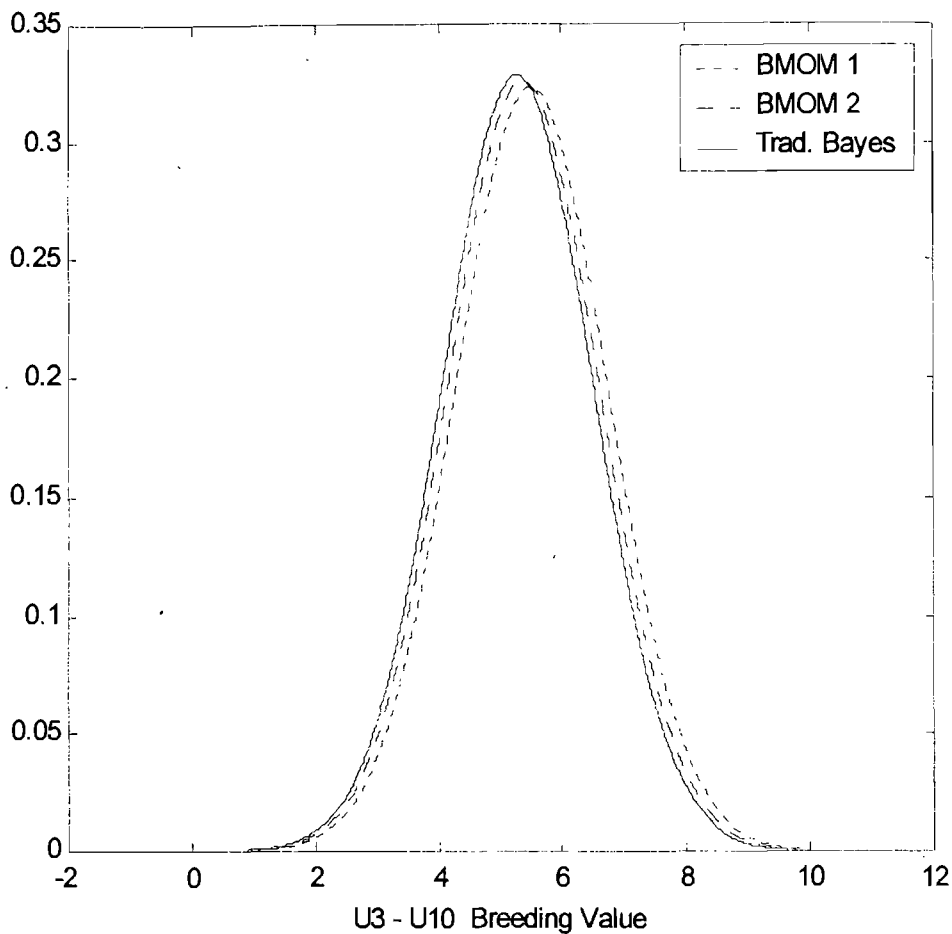


Figure 9. Estimated posterior density of the difference in breeding values between Sire 3 and Sire 10
BMOM 1 (Mean = 5.51), BMOM 2 (Mean = 5.38),
Trad. Bayes (Mean = 5.28)

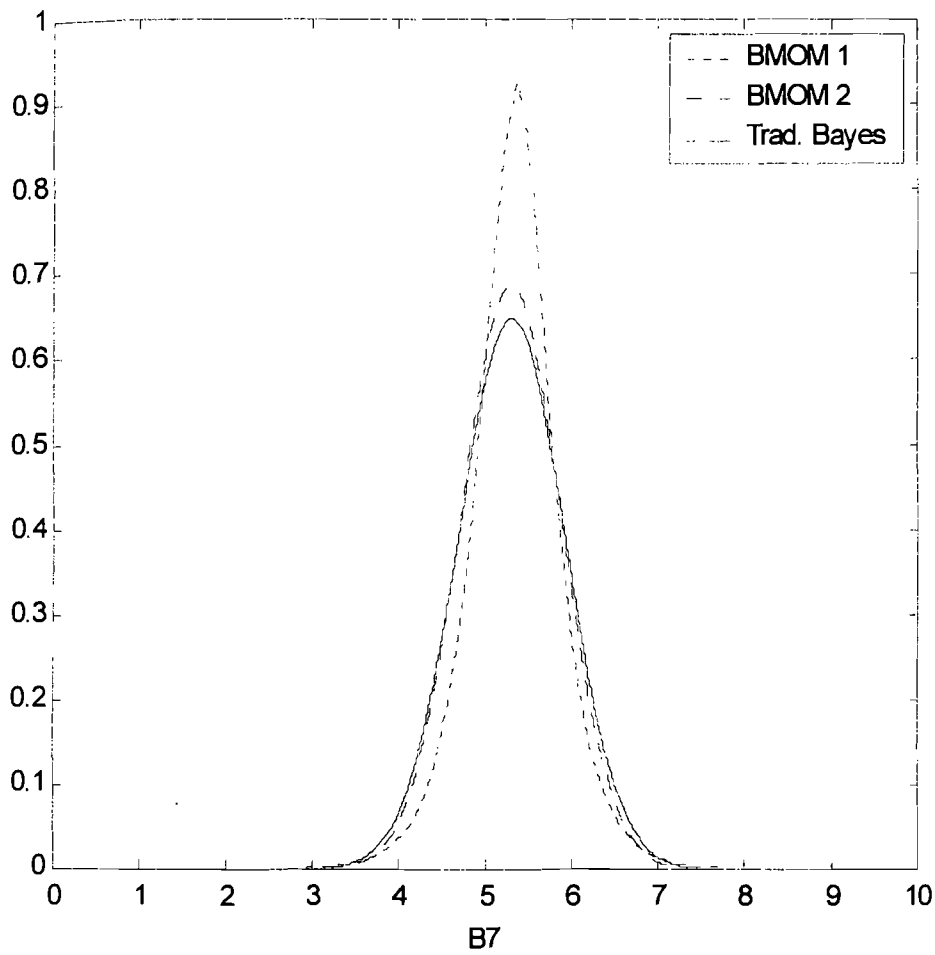


Figure 10. Estimated posterior density of the difference in weaning weights between Year 1 and Year 2 (β_7)

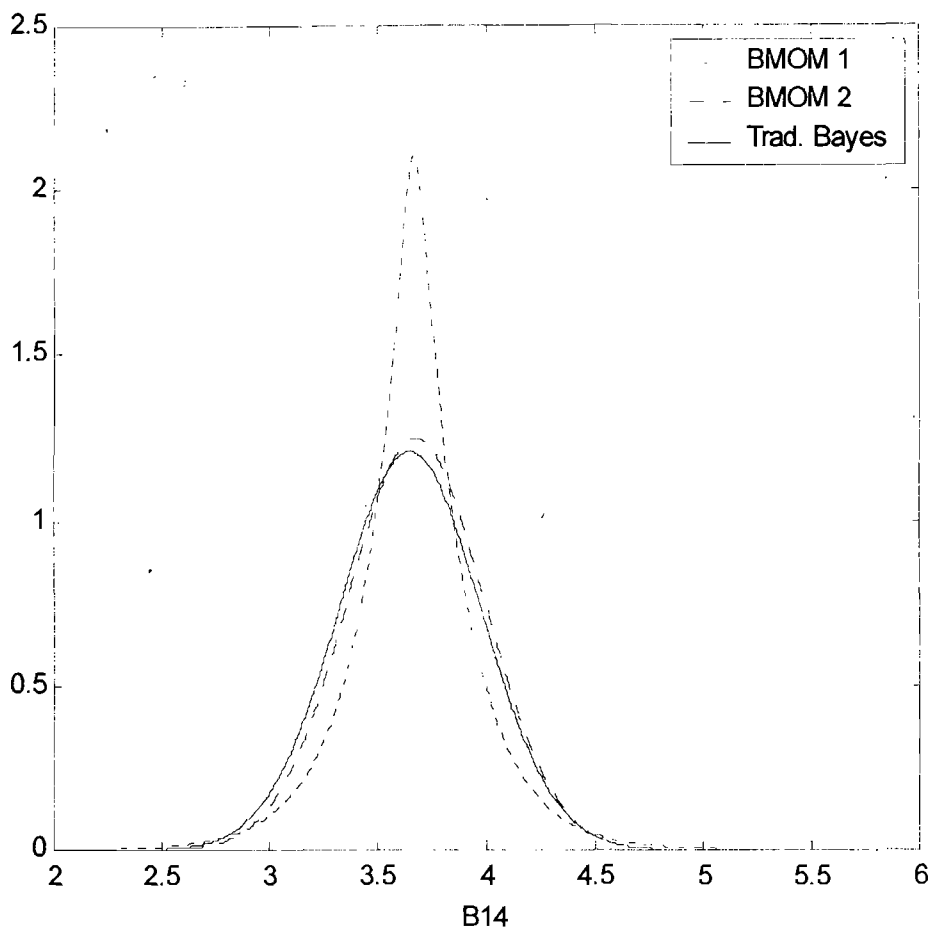


Figure 11. Estimated posterior density of the difference in weaning weights between male and female lambs (β_{14})

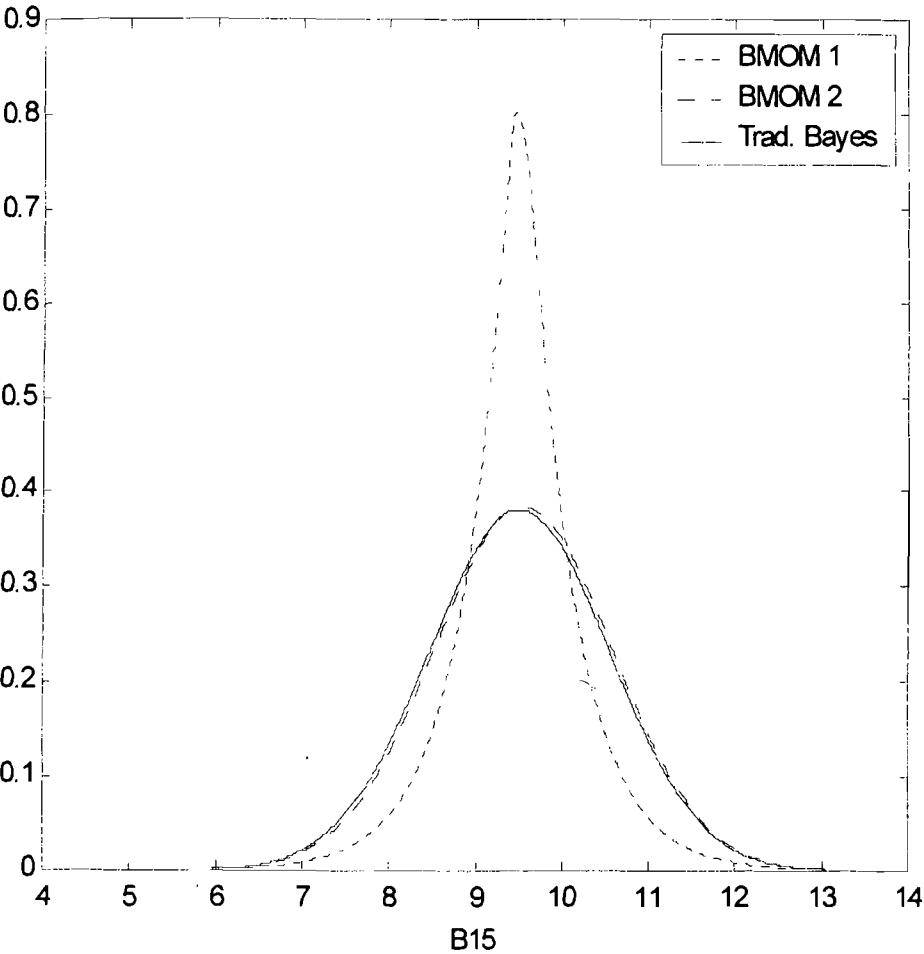


Figure 12. Estimated posterior density of the difference in weaning weights between single births and triplets (β_{15})

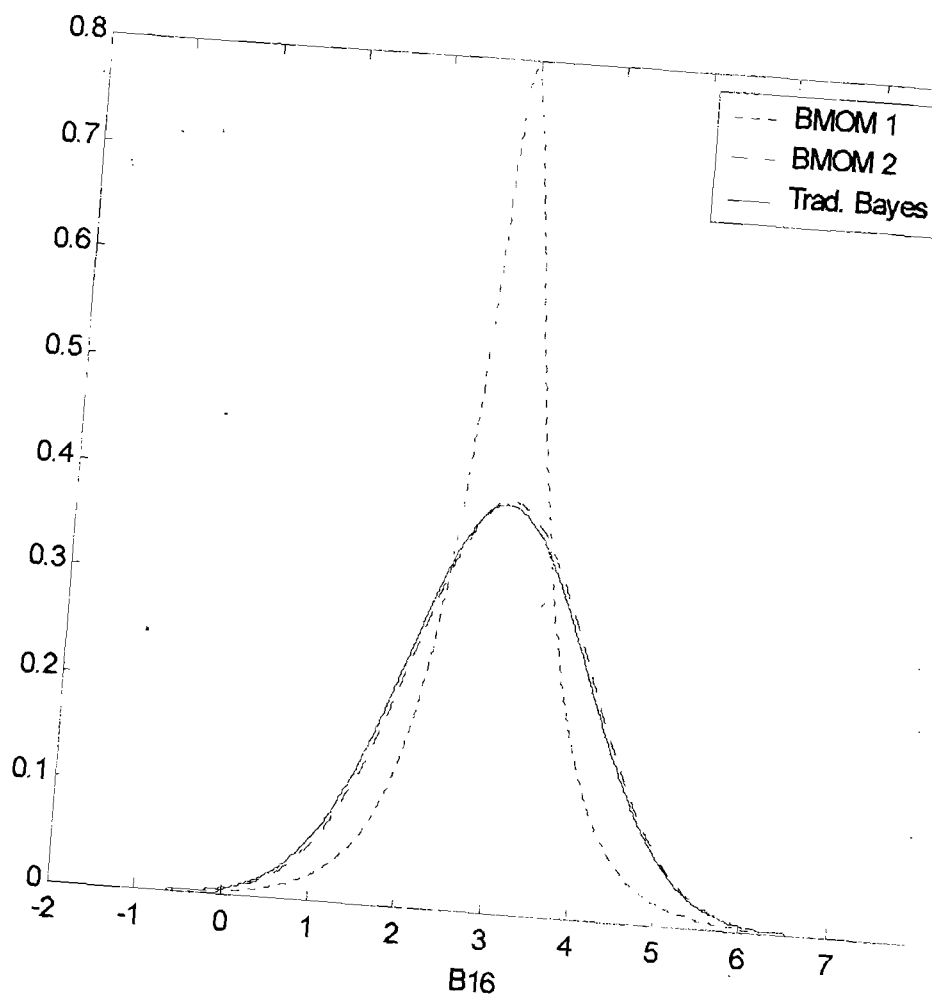


Figure 13. Estimated posterior density of the difference in weaning weights between twins and triplets (β_{16})